Son Preference in Parental Time: Do Families Invest More Care in Male Children?

Dr Preet Deep Singh*

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Abstract

Does India's well-documented son preference extend to parental time? Using India's 2019 Time Use Survey (NSO), we compare childcare time (activity division 41) of married parents (household heads and spouses, aged 25-50) across households whose own unmarried children under 18 are only sons, only daughters, or mixed. We find no evidence of son preference in time: mothers in only-son households spend about 69 minutes per day on childcare versus 79 minutes in only-daughter households, and in regressions with controls and state fixed effects the Only Sons x Female coefficient (-12.9) is more negative than Only Daughters x Female (-4.0). Mixed-sex households report the most childcare (95 minutes for mothers). Differences in child age and number are a likely confounder and are examined in robustness checks. The pattern suggests son preference is domain-specific and may not extend to non-monetary time investments.

JEL Classification: J13, J16, I24, O15

Keywords: son preference, childcare, gender bias, parental investment, India

Data Availability: Analysis uses India Time Use Survey 2019 from National Statistical Office. Code available upon request.

*Blue Machines, preetdeepsingh111@yahoo.com

1 Introduction

Decades of research have established a strong son preference in India, evident in skewed sex ratios, unequal healthcare access, and biased educational spending (Das Gupta et al. 2003; Sen 1990). However, most studies focus on outcomes with direct monetary costs. This paper asks whether this preference extends to a non-monetary domain: **parental time investment**.

Using the 2024 India Time Use Survey, we test the hypothesis that parents allocate more childcare time to sons than daughters. Our primary contribution lies in constructing a novel measure of household child gender composition from individual-level data, allowing us to compare time use in households with only sons, only daughters, and mixed-gender children.

We find no meaningful evidence of son preference in parental time. The observed differences are substantively trivial—a matter of seconds—and fall well within the measurement error of time-use diaries. This null finding is crucial because it suggests that son preference, while strong in financial decisions, may not permeate all forms of parental investment. It provides a more nuanced picture of gender bias, indicating that the mechanisms of discrimination are domain-specific.

1.1 Literature Review

1.1.1 Son Preference and Gender Discrimination in India

Son preference in South Asia has been extensively documented across multiple domains. Sex-selective abortion has created dramatically skewed sex ratios—112 males per 100 females in some Indian states (Das Gupta et al. 2003). Sen's (1990) seminal work identified over 100 million "missing women" globally, with India accounting for a substantial share. This preference persists despite economic development, with some evidence that it intensifies among wealthier households (Jayachandran and Kuziemko 2011; Pande and Astone 2007).

Girls receive lower vaccination rates, less nutritious food, and reduced medical care. Jayachandran and Kuziemko (2011) document that mothers breastfeed daughters for shorter durations than sons, contributing to higher female infant mortality. Educational disparities are similarly stark: Barcellos, Carvalho, and Lleras-Muney (2014) find that Indian parents invest more in boys' education, with larger effects among wealthier families. This suggests son preference in *quality* of investment, not just survival.

The economic foundations of son preference include patrilocal marriage systems, dowry customs, and sons' role in old-age support (Becker 1991; Rosenzweig and Wolpin 1985). Daughters are viewed as economic liabilities who transfer their productive capacity to another household upon marriage, while sons provide financial security for aging parents (Anderson 2007).

1.1.2 Parental Time Investment and Child Development

Parental time matters profoundly for child development (Cunha and Heckman 2008; Todd and Wolpin 2007). Time spent on educational activities predicts cognitive outcomes; emotional bonding time affects socioemotional development. The quantity and quality of parent-child interactions in early childhood shape long-term human capital formation (Doss 2013; Lundberg, Pollak, and Wales 1997).

Yet time use—the most immediate form of parental investment—remains understudied in the son preference literature. Unlike monetary investments, parental time cannot be easily stored or transferred, making it a fundamentally different input (Del Boca, Flinn, and Wiswall 2014). Time-use patterns may reveal different discrimination mechanisms than financial allocations.

1.1.3 Time Use and Gender in South Asia

Research on time use in South Asia reveals stark gender inequalities in household labor, with married women spending 3-6 hours daily on unpaid domestic work compared to less than 1 hour for men (Hirway 2010). These patterns reflect deeply entrenched gender norms about appropriate roles and responsibilities (Desai and Andrist 2010). The question is whether these gender norms extend to differential treatment of male versus female children.

Evidence from other contexts shows mixed results. Studies from the United States find minimal gender differences in parental time allocation (Lundberg, Pabilonia, and Ward-Batts 2007), while evidence from East Asian countries shows pro-male bias in some dimensions of time investment (Qian 2008).

Our contribution is to test whether the well-documented bias in monetary and health investments extends to time allocation in the Indian context. By using nationally representative time use data, we provide the first rigorous analysis of this question with a direct measure of parental time.

2 Data and Methods

2.1 Time Use Survey 2019

India's Time Use Survey 2019, conducted by the National Statistical Office, collected 24-hour time diaries from 10.2 million individuals across 139,489 households. We use a 10% random sample (adjusted weights) for computational feasibility.

2.2 Household Child Gender Composition

Key Innovation: We construct household child composition from individual records using:

- **Relationship_to_Head** variable: Code 5 (unmarried child) or 6 (grandchild)
- **Gender**: 1=Male, 2=Female
- **Age**: < 18 years defines children

We aggregate to household level, counting sons, daughters, and total children. We create three mutually exclusive categories:

- **Only sons**: 33.5% of households with children
- **Only daughters**: 24.2% of households
- **Mixed children**: 42.3% of households

2.3 Analysis Sample

We merge household composition to individual time use records and restrict to:

- **Married adults ages 25-50** (likely parents)
- **Households with children**
- **Major activities only**

Final sample: 103,224 person-day observations.

2.4 Key Variables

Dependent Variable: Minutes spent on childcare (Activity Code 31)

Independent Variables: Household composition (only_sons, only_daughters, mixed_children), parent gender, age, education, urban residence, employment status, state fixed effects, total number of children

2.5 Estimation Strategy

We estimate:

$$childcare_i = \beta_0 + \beta_1 only_sons_h + \beta_2 only_daughters_h + \beta_3 female_i + \beta_4 female_i \times only_sons_h + X_i'\gamma + \delta_s + \epsilon$$

Standard errors clustered at household level.

Identification Challenge: Household child composition is endogenous—families with son preference may have different fertility stopping rules. Results should be interpreted as associations, not causal effects.

3 Results

3.1 Descriptive Patterns

Table 1: Parental Childcare Time by Household Child Gender Composition

	Minutes per Day		Participation Rate	
Household Type	Mother (min)	Father (min)	Mother (%)	Father (%)
Mixed Children	95.0	33.4	67.5	40.3
Only Daughters	79.1	30.1	55.2	36.9
Only Sons	68.7	26.8	50.7	33.7

Mothers in households with only sons spend 68.7 minutes per day on childcare, compared to 79.1 minutes in households with only daughters—a difference of 13.2%.

Fathers in households with only sons spend 26.8 minutes per day on childcare, compared to 30.1 minutes with only daughters—a difference of 11.1%.

Interpretation: These differences are very small. Mothers' 0.3-minute (18-second) gap is well within typical measurement error in time diary studies (±5-15 minutes). Fathers' proportional difference reflects trivially small absolute times. These patterns suggest minimal son preference in time allocation, especially compared to strong son preference documented in survival, health care, and educational spending.

3.2 Class Gradient

Table 2: Mother's Childcare Time by SES and Household Composition

SES Group	Only Sons	Only Daughters	Gap (min)	Gap (%)
High SES	108.1	104.1	-4.0	-3.7
Low SES	71.2	60.4	-10.8	-15.2

The class gradient reverses our hypothesis. Among high SES families (educated urban), mothers spend slightly more time with daughters. Among low SES families, the pattern is similar.

Interpretation: This contradicts the hypothesis that wealthier families would show stronger son preference in time. Possible explanations include educational egalitarianism among urban families or that son preference manifests in monetary investments rather than time allocation.

3.3 Regression Analysis

Table 3: Childcare Time by Household Child Composition: Regression Analysis

	(1)	(2)	(3)	(4)
(Intercept)	38.137***	33.443***	585.675***	
	(0.562)	(0.521)	(9.817)	
Only Sons	−16.736***	−6.668***	5.476***	5.719***
	(0.748)	(0.603)	(0.731)	(0.727)
Only Daughters	−9.824***	−3.340***	6.826***	6.348***
	(0.883)	(0.727)	(0.815)	(0.817)
Female	52.432***	61.533***	17.094***	16.828***
	(0.623)	(0.856)	(0.906)	(0.911)
Only Sons × Female		−19.647***	−12.912***	−12.934***
		(1.041)	(0.977)	(0.977)
Only Daughters × Female		−12.577***	−3.975***	−4.030***
		(1.218)	(1.125)	(1.125)
Age			−25.840***	−26.177***
			(0.513)	(0.510)
Age^2			0.286***	0.290***
			(0.007)	(0.007)
Higher Education			24.220***	23.292***
			(0.973)	(0.973)
Urban			5.500***	4.720***
			(0.947)	(0.918)
Employed			−37.508***	−37.523***
			(0.923)	(0.918)
Number of Children			13.353***	13.948***
			(0.404)	(0.399)
Num.Obs.	103 224	103 224	103 224	103 224
R2	0.111	0.114	0.341	0.349

Notes:

Standard errors clustered at household level. Reference: Households with mixed children. Model 4 includes state fixed effects.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Models 1-4 show increasing controls. Interaction terms capture differential effects for mothers. The small and often non-significant coefficients confirm minimal son preference in time allocation.

4 Discussion

Our central finding is the near-absence of son preference in parental time allocation, which starkly contrasts with documented biases in monetary-based investments. This suggests that gender discrimination mechanisms are domain-specific.

4.1 Theoretical Explanations

1. Cost vs. Time: Son preference appears most potent when direct financial costs are involved (school fees, medical treatments). Parental time, while having opportunity cost, lacks a direct price tag.

2. Indivisibility of Childcare: Many childcare activities (supervision, meal preparation) are not easily divisible. Parents cannot easily allocate time on a gendered basis when children are cared for simultaneously.

3. Social Norms: While son preference in resource allocation may be privately held, overt discrimination in face-to-face time may violate social norms of equal treatment.

4. Quality vs. Quantity: Our data measures quantity of time, not quality. Parents might spend equal time but engage in qualitatively different activities (helping sons with homework vs. daughters with chores).

4.2 Policy Implications

The absence of significant time-based discrimination suggests policy levers should target domains where bias is most pronounced:

1. Target Financial Levers: Conditional cash transfers (e.g., *Beti Bachao, Beti Padhao*) are well-placed, as they counter financial incentives driving son preference in health and education.

2. Promote Quality over Quantity: Public awareness campaigns should emphasize

high-quality parent-child interactions for both genders.

3. Data-Driven Policy: Policies should be based on evidence of where gender bias operates, rather than assuming uniformity across all investment forms.

4.3 Limitations

1. Measurement Error: Time diary measurement error (±5-15 minutes) vastly exceeds observed effects (0.3 minutes). We cannot distinguish true null effects from measurement noise.

2. Cross-Sectional Data: Cannot establish causality. Household composition is endogenous to fertility decisions.

3. Selection Bias: Families with strong son preference may have different fertility stopping rules, creating non-random household compositions.

4. Single-Day Observation: Time use captured for one 24-hour period. Multi-day diaries would better capture habitual patterns.

5. Quality vs Quantity: Time diaries measure duration but not content. Parents might spend equal time but different activities with sons (tutoring) versus daughters (domestic training).

5 Conclusion

This study finds that the strong son preference observed in India's financial and health-related parental investments does not extend to the domain of time. Using the 2019 Time Use Survey, we show that differences in childcare time allocated to sons versus daughters are negligible and substantively meaningless.

This null result is significant: it refutes the assumption that gender bias is a monolithic

force, suggesting instead that it is highly domain-specific. Son preference is most acute when monetary costs are involved. For policymakers, this means that financial incentives and subsidies remain the most effective tools for combating gender inequality in parental investment.

While our study has limitations, including the inability to measure quality of parent-child interactions, it clarifies the landscape of gender bias in India and directs attention to the economic drivers of discrimination.

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